Ziting (Kay) Tang

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Research Interests: Machine learning for high-dimensional biomedical data; multimodal analysis of combined imaging and omics data, with a current focus on cancer

EDUCATION

Rice University Houston, TX | Aug 2022 - May 2024

Master of Statistics | GPA 3.8/4.0

- Machine Learning & AI Courses: Neural Machine Learning I, Data Science & Machine Learning, Introduction to Computer Vision, Introduction to Information Retrieval
- Statistics & Mathematics Courses: Probability, Mathematical Probability I, Statistical Inference, Regression and Linear Models, Multivariate Analysis, Advanced Statistical Methods
- Computing Courses: Graduate Design and Analysis of Algorithms, Computer Systems Architecture, Big Data Management for Data Science, R for Data Science
- Independent Research (STAT 590): Graph similarity measures and visualization for connectivity graphs of Self-Organizing Maps (Advisor: Dr. Erzsébet Merényi)

Zhongnan University of Economics and Law Wuhan, China | Sep 2016 - Jul 2020

Bachelor of Science in Statistics | GPA 3.2/4.0

- Thesis: Graph-based Collaborative Filtering for Recommendation (Advisor: Dr. Yuanshan Wu)

PUBLICATIONS & PRESENTATIONS

- **Tang, Z.**, Enriquez, J. S., Dutta, P., Bhattacharya, P. K., and Merényi, E. "Characterization and Correction of Combined Urea and Noise Influence in Hyperpolarized ^{13}C Pyruvate Magnetic Resonance Spectroscopy." Manuscript in preparation.
- **Tang, Z.**, Enriquez, J. S., Dutta, P., Bhattacharya, P. K., and Merényi, E. "Correction of Combined Urea and Noise Influence in Hyperpolarized Metabolic Imaging." *2027 ISMRM-ISMRT Annual Meeting*, Vancouver, BC, Canada, May 8-13, 2027. Submitted.
- Merényi, E., **Tang, Z.**, and Mendenhall, M. J. "Characterization of SOM Model Quality with Graph Property Measures." *16th International Workshop on Self-Organizing Maps, Learning Vector Quantization & Beyond (WSOM+ 2026)*, Manchester, UK, Nov 19-21, 2026. In press.
- **Tang, Z.**, Enriquez, J. S., Dutta, P., Bhattacharya, P. K., and Merényi, E. "Correction of Combined Urea and Noise Influence in Hyperpolarized Metabolic Imaging." Poster presented at the *15th Annual Metabolism in Cancer Symposium*, MD Anderson Cancer Center, Houston, TX, Sep 17, 2026.
- Merényi, E., Bhattacharya, P. K., Pilié, P. G., Enriquez, J. S., Dutta, P., **Tang, Z.**, and Cardenas, C. "Delineating Dynamic Hyperpolarized Metabolic Signatures in Patient-derived Prostate Cancer Xenografts Using AI." *2026 ISMRM-ISMRT Annual Meeting*, Cape Town, South Africa, May 9-14, 2026. Program #662-04-001.

RESEARCH AND WORK EXPERIENCE

Rice University, Department of Statistics Houston, TX | May 2025 - present

Research Technician II

Supervisors: Drs. Erzsébet Merényi (Rice University) and Pratip K. Bhattacharya (The University of Texas MD Anderson Cancer Center)

Project: AI for Early Identification of Androgen-Indifferent, Lethal Prostate Cancer and Therapy Selection, funded by the Cancer Prevention and Research Institute of Texas (CPRIT) grant RP250090. PI: E. Merényi, Rice University.

- Built a pipeline that digests raw hyperpolarized MR spectroscopy data cubes from mice bearing patient-derived xenograft (PDX) tumors and distills them into metabolic signatures ready for input to machine learning. Adapted the same pipeline for NMR metabolomics data, with flexibility to apply to other omics data.
- Developed an approach to correct artifacts in hyperpolarized MR spectra, which improved quantification of pyruvate-to-lactate conversion and enabled reliable inclusion of previously unusable data cubes (presented at MICS 2026; ISMRM 2027 abstract submitted; manuscript in preparation).
- Produced combined views of metabolic signatures, relevant metadata, and clustering results, to help facilitate identification of metabolically distinct prostate cancer subgroups (ISMRM 2026).

Rice University, Department of Statistics Houston, TX | Jan 2024 - May 2025

Research Assistant

Supervisor: Dr. Erzsébet Merényi (Rice University)

Project: Learning More from Observations by Integrating Multiple Perspectives, funded by AFOSR grant FA9550-23-1-0642. PI: E. Merényi, Rice University.

- Surveyed graph measures that characterize the knowledge learned by Self-Organizing Maps (SOMs) through the so-called connectivity (CONN) graph, an advanced knowledge representation; led the summarization of over 30 measures we investigated.
- Generated extensive Excel tables of these measures for different SOM states of various data sets, for subsequent systematic evaluation of the measures' ability to identify the known best SOM state of each data set.
- Helped design vertex-level visualizations of proposed topology-violation and entropy measures, layered with the CONN graph on the SOM, for insightful representations of the contributions of individual SOM prototypes to model reliability (WSOM+ 2026).
- Built a repeatable, automated pipeline that mass-produces Excel sheets/tabs of several variants of the above graph measures, and maps, from series of SOM states.

Wisers Information Limited Shanghai, China | Jun 2021 - Nov 2021

Data Analyst Intern

- Produced daily and weekly market reports on newly launched client products, applying frequency and sentiment analysis in Python to 100k+ keyword-related user posts from mainstream social media platforms.
- Collaborated on building an NLP filtering and processing system for large-volume raw text, streamlining data cleaning for downstream analysis and improving analysis-cycle efficiency by 12%.
- Benchmarked client products against competitors and across release periods using platform-specific Target Group Index (TGI) analysis.

TEACHING EXPERIENCE

Teaching Assistant, Rice University Houston, TX | Jan 2023 - Dec 2024

Neural Machine Learning I (Spring 2024, Fall 2024)

- Held office hours on neural machine learning methods and programming, graded assignments and exams with feedback, and assisted with classroom management.

R for Data Science (Spring 2023, Fall 2023)

- As part of a teaching assistant team for a large course, graded coding assignments and exams and helped students with data analysis in R in one-on-one sessions.

SKILLS

Programming: R, MATLAB, Python (PyTorch, scikit-learn, OpenCV), SQL (PostgreSQL), Linux/Bash, Git
Big Data & Cloud: Apache Spark (PySpark), AWS (EMR, S3)
Other Software: LaTeX, Microsoft Office, Visio
Languages: Mandarin (Native), English (Proficient), Korean (Intermediate)
Personal: Vocalist and hip-hop dancer